Aaradhya Dev Tamrakar

Applied Machine Learning · AI Workflows · Model Evaluation & MLOps

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Final-year engineering student and Fuse AI Fellow building applied machine learning pipelines, model-evaluation frameworks, and developer automation tools. Experienced in demographic performance auditing (AIF360, Fairlearn), tabular classification and explainability (XGBoost, SHAP), and containerized API services with FastAPI and Docker. Focuses on turning repetitive development steps into reliable tools and reproducible evaluation checks.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics, Communication and Information Engineering | December 2022 – Expected January 2027

Kathmandu Engineering College, Institute of Engineering (IOE), Tribhuvan University • Cumulative Average: ~75% across completed semesters

Relevant Coursework: Machine Learning · Signals and Systems · Digital Electronics · Control Systems · Communication Systems · Engineering Mathematics · Probability and Statistics · Object-Oriented Programming (C++) · Computer Networks · Microprocessors

FELLOWSHIPS & APPLIED ML EXPERIENCE

Fuse AI Fellow — Fusemachines AI Fellowship 2026 | 2026 – Present (Active)

- Co-developed **BiasAperture**, an automated demographic fairness auditing pipeline measuring predictive disparities on computer vision benchmarks as the fellowship capstone project.
- Built applied agentic and ML coursework pipelines including a 5-stage Text-to-SQL system (Planner → Generator → Validator → Executor → Summarizer) tested against structured evaluation suites.
- Developed customer churn and lifetime value (CLV) prediction workflows utilizing **SMOTE**, cost-sensitive threshold tuning, and **SHAP** explainability.
- Implemented MLOps coursework for experiment tracking, containerized microservices (**FastAPI**, **Docker**, **PostgreSQL**), and scheduled data-drift evaluation.
- Evaluated probabilistic estimation and time-series forecasting ensembles on retail datasets as part of advanced predictive modeling modules.

NSSR DataCamp Fellow (Cohort 2) — Nepalese Society of Student Researchers | April 2026 – Present

- Selected in competitive cohort for sponsored DataCamp access; developed applied GenAI workflows automating data cleaning, schema normalization, and exploratory analysis.
- Completing advanced technical pathways in SQL database design, Python data analysis, and predictive modeling.

RESEARCH & TECHNICAL PROJECTS

BiasAperture — Computer Vision Demographic Fairness & Bias Audit | 2026

Python · AIF360 · Fairlearn · Statistical Testing · Jinja2 · OpenCV

- Co-developed an automated prediction-auditing and reporting pipeline as a two-person fellowship capstone with Tisha Manandhar.
- Audited classification performance across **126 demographic bins** on the FairFace benchmark (**97,698 images**), computing Disparate Impact Ratio, Demographic Parity Difference, and Equalized Odds.
- Engineered self-contained HTML and PDF report generation with confidence intervals and diagnostic disparity plots; matched model comparison and validation remain active work.

SPARK — Wearable Edge-AI Fall Detection System | 2025 – Present

ESP32-S3 · C/C++ · TensorFlow Lite Micro · Python · FreeRTOS · MPU-6050

- Project lead for ML modeling, gateway processing, and explainability in a four-person team; hardware and firmware have dedicated owners.
- Trained 1D-CNN fall-detection models on the SisFall benchmark (38 subjects, 38,420 windows), evaluating subject-stratified cross-validation with **0.9185 AUC-ROC** and 87.81% sensitivity.
- Quantized model to an **18.5 KB INT8 CNN** footprint for ESP32-S3 edge deployment; on-device hardware testing, BLE streaming, and volunteer cohort validation remain pending.

Fleet Orchestrator — AI Workflow Coordination & Tooling | 2025 – Present

Python · FastAPI · Provider Adapters · Model Context Protocol (MCP) · SQLite

- Built coordination tooling for multi-step AI workflows, with task state, worker adapters, scheduling, and live status inspection.
- Standardized provider adapters (local models, CLI harnesses, API endpoints) with reusable interfaces to eliminate repeated orchestration scripts.

nepali-ocr-ai — Low-Resource Document Processing Pipeline | 2025

Python · Tesseract OCR · OpenCV · Devanagari NLP

- Prototyped a document intelligence pipeline combining OCR with Devanagari text normalization and Preeti ↔ Unicode font transcoding.
- Built preprocessing filters in OpenCV for uneven illumination and historical print noise reduction.

LEADERSHIP & SERVICE

Vice Chair (2026 – Present); Vice Secretary (2025 – 2026) — IEEE KEC KTM Student Branch | 2025 – Present

- Coordinate branch planning, technical workshops, and cross-committee communication for Kathmandu Engineering College IEEE student body.
- Facilitated technical awareness events and student engagement initiatives across engineering departments.

Event Manager (2026 – Present); Resource Manager (2024 – 2026) — Electronics Project Club (EPC), Kathmandu Engineering College | 2024 – Present

- Manage event logistics, hardware inventory, and project exhibitions for the college electronics department.
- Organized project showcases and hardware component distribution for student engineering competitions.

TECHNICAL SKILLS

Machine Learning & Evaluation: Python, PyTorch, scikit-learn, Pandas, NumPy, XGBoost, SHAP, Fairlearn, AIF360, OpenCV, MediaPipe

AI Systems, Backend & Tooling: FastAPI, Docker, Model Context Protocol (MCP), SQL (PostgreSQL, SQLite), asyncio, REST APIs, MLflow

Engineering Practice & Tooling: Git, GitHub, Linux, PowerShell, Pytest, Reproducible Configurations, WordCraft, LaTeX